

Agent 1: empirical dimensionality reduction

BIOS 774 · Local reproducible statistical agent · Scientific wording is a draft for user review.

Problem and contribution

A useful dimensionality-reduction workflow should explain why a method advances on the dataset at hand. This agent inspects numerical data, chooses explicit preprocessing, attempts all supported methods on a common pilot, and uses measured quality, stability, failures, and runtime to select a small final candidate set. It then tunes only that shortlist and produces auditable reports.

Architecture and tools

A Python src package exposes inspect → pilot → select → tune → final → report → validate through one CLI and run-all. Typed configuration, atomic JSON artifacts, content checksums and subprocess isolation make stages resumable. SLURM supplies CPU execution; Python scientific packages implement numerical tools. Jinja2 and WeasyPrint generate PDFs, checked with pypdf. No multi-agent framework, web service or database is required.

All ten methods are real implementations: PCA, Kernel PCA, MDS, Isomap, LLE, Laplacian Eigenmaps, Diffusion Maps, Bayesian GPLVM, t-SNE and UMAP. Scikit-learn supplies seven, pydiffmap supplies diffusion maps, GPyTorch optimizes Bayesian GPLVM, and umap-learn supplies UMAP. The GPLVM loss and changes from PCA initialization are recorded.

Dataset-aware planning and decision logic

PBMC count data undergo fixed cell QC, cohort-specific gene filtering, library normalization, log transformation, variable-gene selection, and scaling. PathMNIST uses consistently scaled RGB pixels. Cohorts are stratified where annotations exist, while all fitting and selection exclude labels. A cohort-fitted PCA representation supplies identical numerical input to every method.

Quality combines trustworthiness (weight 0.45), nearest-neighbor recall (0.45), and seed stability (0.10). Methods meeting 90% of the best quality or on the quality-runtime Pareto frontier are eligible; candidates are ranked by quality, runtime, then name. Successful PCA is retained as the baseline, with at most four finalists. Severe connectivity or singularity warnings exclude a method. All original warnings and failures remain visible.

Stochastic methods have repeated seeds; deterministic stability is defined as one. Shortlist refinement uses the same criteria. Silhouette is secondary. These are descriptive in-cohort analyses; there is no held-out inference claim.

pbmc3k: evidence and outcome

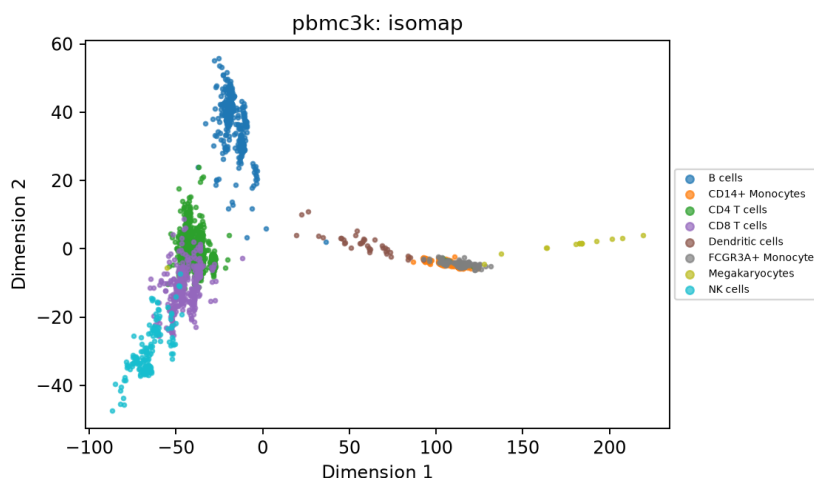
Source observations: 2700; source features: 32738; pilot cohort: 1000; final cohort: 2638. Post-QC available cells: 2638.

Method	Status	Quality	Seconds	Decision
pca	success	0.486	1.81	retain
kernel_pca	success	0.453	1.89	exclude
mds	success	0.415	8.06	exclude
isomap	success	0.501	2.03	retain
lle	success	0.463	1.80	exclude
laplacian	success	0.496	1.74	retain
diffusion_maps	success	0.456	2.82	exclude
gplvm	success	0.418	16.49	exclude
tsne	success	0.595	7.64	retain
umap	success	0.483	24.10	exclude

Retained: pca, tsne, isomap, laplacian. Quality threshold: 0.5354. Detailed reasons distinguish method priors from the empirical outcome in decisions.json.

Final method	Trustworthiness	Neighbor recall	Silhouette
pca	0.825	0.051	0.404
tsne	0.897	0.208	0.419
isomap	0.822	0.051	0.329
laplacian	0.838	0.077	0.390

The reported final metrics quantify local geometry preservation on the declared final cohort. They should not be directly compared with pilot metrics as if the observation sets were identical. Full tuning evidence, parameter choices and final embeddings accompany the dataset report.



Example final embedding. Colors are diagnostic annotations; apparent separation alone does not demonstrate biological validity.

pathmnist: evidence and outcome

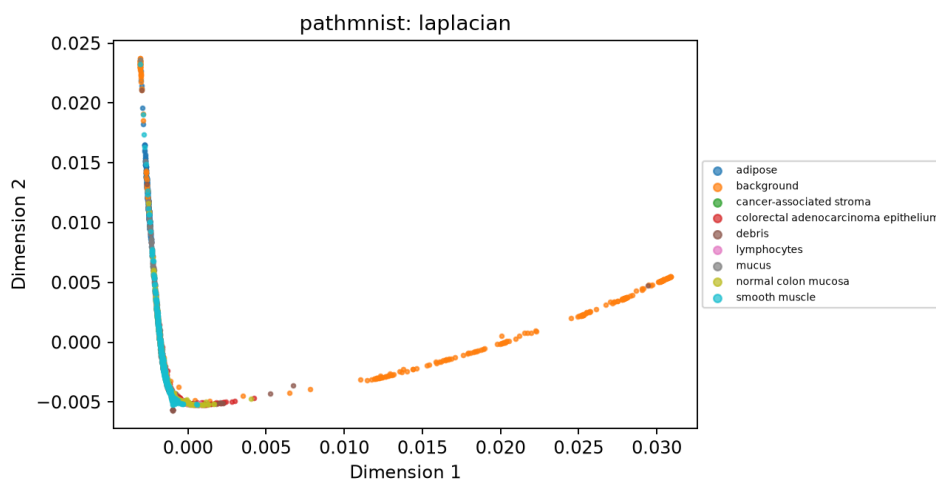
Source observations: 89996; source features: 2352; pilot cohort: 1000; final cohort: 3000. **The final analysis is a stratified train-split subset, not the complete PathMNIST dataset.**

Method	Status	Quality	Seconds	Decision
pca	success	0.587	1.63	retain
kernel_pca	success	0.560	3.00	exclude
mds	success	0.554	2.45	exclude
isomap	success	0.567	1.68	exclude
lle	success	0.557	1.51	retain
laplacian	success	0.564	1.60	retain
diffusion_maps	success	0.556	1.98	exclude
gplvm	success	0.479	11.42	exclude
tsne	success	0.649	5.76	retain
umap	success	0.573	19.80	exclude

Retained: pca, tsne, laplacian, lle. Quality threshold: 0.5845. Detailed reasons distinguish method priors from the empirical outcome in decisions.json.

Final method	Trustworthiness	Neighbor recall	Silhouette
pca	0.854	0.151	-0.101
tsne	0.881	0.244	-0.086
laplacian	0.843	0.119	-0.148
lle	0.832	0.097	-0.121

The reported final metrics quantify local geometry preservation on the declared final cohort. They should not be directly compared with pilot metrics as if the observation sets were identical. Full tuning evidence, parameter choices and final embeddings accompany the dataset report.



Example final embedding. Colors are diagnostic annotations; apparent separation alone does not demonstrate biological validity.

Assessment and reproducibility

Strengths and limits

The same public interface adapts to sparse count data and dense images, with transparent common-cohort comparisons and quantitative candidate selection. Traceable artifacts preserve failed attempts and numerical warnings instead of presenting only attractive embeddings. Runtime-aware cohort planning makes expensive methods feasible under a declared budget.

Results depend on representation, metric weights, neighborhood scale and random seeds. Local metrics do not fully measure global structure. Deterministic methods receive a defined stability value; this is not proof of robustness to resampling. GPLVM uses a fixed optimization budget and may need longer training. Disconnected graph warnings conservatively exclude methods. Cold-start imports affect runtime, and cubic planning bounds are conservative assumptions rather than measured scaling laws.

PathMNIST pixel distances need not reflect all histological semantics. PBMC processed labels originate from prior clustering and do not provide independent biological ground truth. No causal, clinical or generalization conclusion follows from the embeddings. Neither source is sent to an external model in deterministic mode.

Reproduction and audit

Run `scripts/01_setup_environment.sh` for the exact agent environment, then `scripts/03_smoke_test.sh` inside a SLURM allocation. Stage scripts in `scripts/slurm` accept a dataset config. Run `python -m agent1_dr run-all --config config/pbmc3k.yaml --resume` and the analogous PathMNIST command on compute nodes. Finally run `project-report` and `validate --all-reports`. The lock files, configuration/code hashes, source checksums, cohort IDs, seeds, and job manifests specify the evidence chain.

Three PDFs are generated from validated artifacts. The default mode is deterministic templating, not an LLM-written report. An optional OpenAI interface can organize exact evidence sentences and rejects unsupported claims; live API operation requires credentials. Tests exercise the actual methods, GPLVM optimization, failure isolation, metric behavior, selection and a fresh end-to-end synthetic run.

References

10x Genomics, PBMC3k public dataset; Scanpy dataset documentation, <https://scanpy.readthedocs.io/>. Yang et al. (2023), MedMNIST v2, Scientific Data 10, 41, doi:10.1038/s41597-022-01721-8. Titsias & Lawrence (2010), Bayesian Gaussian Process Latent Variable Model, AISTATS. GPyTorch GPLVM SVI tutorial, <https://docs.gpytorch.ai/>. Scikit-learn, pydiffmap and umap-learn official documentation. Source and implementation links: `docs/DATA_PROVENANCE.md` and `docs/METHOD_NOTES.md`.